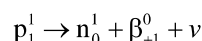


1. β^- emission from the nucleus is always accompanied by an antineutrino according to nuclear reaction : $n \rightarrow p + e^- + \bar{\nu}$ [antineutrino]. Energy of α -particle emitted from a given nucleus is always constant and is equal to

$$\frac{A}{A-4}Q. \gamma \text{ emission takes place while an excited}$$

nucleus goes to ground state and thus becoming more stable.

2. Neutron is the most effective projectile as it can hit the nucleus directly without being attracted or repelled by nucleus or electrons. Tritium is not found in observable quantity as its half life is shorter than life of universe. A free proton cannot convert into a neutron but a proton inside nucleus can convert according to the reaction :



3. A good moderators has two properties; it slows down neutrons by elastic collision and it doesn't remove them from the core. That is why lighter elements are better moderators. Heavy water is used in reactors because it has lesser absorption probability of neutrons than ordinary water. Mass defect does take place in chemical processes. However, the mass defects involved in chemical reactions are almost negligible than those in a nuclear reactions. Cadmium rods are used for absorbing thermal neutrons and keep chain nuclear reaction controlled.
4. Mass spectrometer is used to measure atomic mass. Chemical properties depend upon the electronic configuration and isotopes have same electronic structure, hence isotopes have identical chemical behavior. Atomic mass is the weighted average of atomic masses of different isotope of same element takes against relative abundance, therefore it may come out to be non-integer.

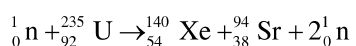
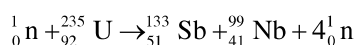
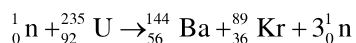
However, from bombardment of beryllium nuclei with α particles neutral radiations consisting of neutrons are emitted that have much lower energy than photons.

5. Electron scattering gives more accurate results because electron scattering senses charge distribution while α scattering senses nuclear matter. If α -particle of higher energy are used, they may come closer and suffer strong short ranged nuclear forces. However, Rutherford took only columbic forces into account. Deviation of an α -particle is governed by impact parameter-lesser the impact parameter, higher the deviation. However, α -particles suffer elastic scattering against gold nuclei because neither the α particle nor the gold nuclei are internally excited during collision.
6. The property that a given nucleon influences only nucleons close to it makes is referred as saturation property of nuclear force. Inside nucleus, the order of strength is : $F_N > F_E > F_G$. However, potential energy associated with nuclear force is minimum but non-zero when distance between nucleons is 0.8 fm
7. Radioactivity was discovered by Henry Becquerel. A spontaneous decay is an exothermic process, where energy liberation is due to mass defect as per $E = \Delta mc^2$. According to Rutherford - Soddy law, radioactivity is governed by law of probability where we cannot predict which radionuclide disintegrates first. Out of three Uranium isotopes i.e., ${}_{92}\text{U}^{233}$, ${}_{92}\text{U}^{235}$ and ${}_{92}\text{U}^{238}$, the highly abundant ${}_{92}\text{U}^{238}$ is non-fissionable.
8. β^- is generated when neutron converts into proton $n \rightarrow p + \beta^- + \bar{\nu}$ while β^+ is emitted when proton converts into neutron $p \rightarrow n + \beta^+ + \nu$. Neutrinos are neutral particles, have almost negligible [probably zero] mass as compared to electrons and penetrate deep into matter while showing weak interaction with other particles. This makes them hard to detect.
9. Two nuclei undergoing fusion must come close enough so that attractive short- range nuclear force is able to affect them. However, since they are both positively charge particles, they experience columbic repulsion. They, therefore, must have enough energy to overcome columbic barrier which is provided by raising the temperature that is in the order of 10^8 K. Thermonuclear fusion of hydrogen is the source of energy output inside stars. Once the hydrogen depletes, starts starts collapsing under gravity, which will raise the temperature of core and star starts expanding resulting in red giant star.

10. Greater the binding energy, the less is the total mass of the bound system, such a nucleus. Therefore, energy is released when nuclei with less total binding energy transform to nuclei with greater binding energy. A chemical equation is balanced in the sense that the number of atoms of each element is the same on both sides of the equations. In a nuclear reaction, elements may be transmuted. Thus, the number of atoms of each element is not necessarily conserved in a nuclear reaction. However, the number of protons and the number of neutrons are both separately conserved in a nuclear reaction. Now, since proton number and neutron number are conserved in a nuclear reaction, the total rest mass of neutrons and protons is the same on either side of a reaction. But the total binding energy of nuclei on the left side need not be the same as that on the right hand side. The difference in these binding energies appears as energy released or absorbed in a nuclear reaction. Since binding energy contributes to mass, the difference in the total mass of nuclei on the two sides get converted into energy or vice-versa.

However, in β^- decay an antineutrino is generated when neutron converts into proton [$n \rightarrow p + \beta^- + \bar{\nu}$] while in β^+ decay a neutrino is generated when proton converts into neutron [$p \rightarrow n + \beta^+ + \nu$].

11. Energies involved in exothermic chemical reactions are in the range of eV, while in nuclear reactions energy release is of order of MeV. Therefore, fission of 1 kg of Uranium produces 10^{14} J, while burning of 1 kg coal generates 10^7 J. Fission of ${}_{92}^{235}\text{U}$ may result in different pair of daughter nuclei :



But approximately same amount of energy [200 MeV] is released per fission. When the organism is dead, its interaction with the atmosphere (which maintains the above equilibrium activity) ceases and its activity begins to drop. From the known half-life (5730 years) of ${}_6^{14}\text{C}$, and the measured activity, the age of the specimen can be approximately estimated

12. In some reactions, 2 neutrons are emitted while in some 3 neutrons are emitted, making an average of $2\frac{1}{2}$ neutrons per fission. These neutrons, unless slowed down, will escape from the reactor without interacting with the uranium nuclei. Therefore, moderators are provided to slow down thermal neutron via elastic collisions. However, fast breed reactors use fast neutrons for sustaining chain nuclear reaction, hence moderators are not required.
13. Nuclei also have energy level like atoms. Atomic energy level spacing is of the order of eV, while difference in nuclear energy levels is MeV. By electron scattering experiments, it was found that radius of nuclei, $R = R_0 A^{1/3}$ and also $M = mA$, where m is the mass of one nucleon. This implies that nuclear density is independent of A . In heavier nuclei, more neutrons are required to overcome the effect of repulsion among the protons, hence neutron : proton ratio increases slightly to 3 : 2. However, inside nucleus a proton may convert into neutron by reaction $p \rightarrow n + \beta^+ + \nu$
14. After α or β decay nucleus goes into excited state and spontaneously decays into ground state emitting a γ particle with energy equal to energy difference between two states. Electrons and positrons can undergo pair annihilation reaction : $e^- + \beta^+ \rightarrow 2\gamma$. However, if β^+ emission is energetically allowed, electron capture is necessarily allowed but vice versa is not true.
15. Q-value i.e., energy liberated during nuclear reactions appear as kinetic energies of products of which kinetic energy of daughter nuclei is

$$KE_D = \frac{4}{A}Q, \text{ while kinetic energy of } \alpha\text{-particle}$$

is given by $KE_\alpha = \frac{A-4}{A}Q$. In both β^+ and β^- decay, mass number remains unchanged, hence reactant nuclei and product nuclei are isobar. However, in β -decay continuous spectrum is observed.

16. As we know that,
 α -particle: 2 unit of positive charge
 β -particle: 1-unit positive charge
 γ -particle: No charge
 Due to this reason, energy levels of the atom change for α and β but not for others. So the electronic energy level of the atom changes in emission of α and β particle, but not in γ decay.

17. Electrostatic force between proton-proton is repulsive which causes the instability of nucleus. So, neutrons are larger than proton.
18. Half-life time for a radio-active substance is defined as the time in which a radioactive atomic substance remains half of its original value of radio-active atom. Given that, Half-life = 1 year. So, after 1 year means one half-life, that is, average atoms of radioactive substance remain after 1 year in each container is equal to half of $50,000 = 25,000$ atoms (average).
19. For elastic collision masses of both must be equal so that they can exchange the velocities. To slow down the speed of neutron substance should be made up of 1 proton for perfectly elastic, that is, we need light nuclei not heavy. For heavy nuclei only direction will change not the speed.
21. For fusion to take place, the 2 nuclei must come close enough so that attractive short-range nuclear force is able to attract them. However, since both are positively charged particles, they experience coulomb repulsion. To overcome this repulsion they must have enough kinetic energy. This is achieved by high temperature.